

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA5116 COURSE NAME: Cryptography and Network Security

COURSE OUTCOME COVERED

CO3: *Examine cryptographic hash algorithms, digital signature schemes, and assess Kerberos and X.509 authentication frameworks for ensuring integrity, authentication, and non-repudiation.CO (BL4 , BL5)*

Assessment Tool Weightage: 10%

QUESTIONS: 5

TOTAL MARKS: 50

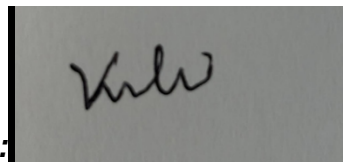
Annexure A MCQ

Code of Conduct:

I _____VARSHA.S/192511358_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A rectangular box containing a handwritten signature in black ink. The signature appears to be 'V. S.' or similar, written in a cursive style.

Annexure A

MCQ

MCQ (Additional Set)

1. Which property ensures the same input always produces the same hash?
 - A) Avalanche effect
 - B) Determinism
 - C) Collision resistance
 - D) Diffusion
2. What is the primary purpose of SHA-3?
 - A) Faster encryption
 - B) Alternative secure hashing standard
 - C) Data compression
 - D) Key exchange
3. Which attack tries all possible inputs to match a hash?
 - A) Replay attack
 - B) Brute-force attack
 - C) MITM attack
 - D) DoS attack
4. What is a salted hash?
 - A) Encrypted hash
 - B) Hash with random data added
 - C) Shortened hash
 - D) Reversed hash
5. Which algorithm is widely used for password hashing?
 - A) AES
 - B) RSA
 - C) bcrypt
 - D) DES
6. What does second pre-image resistance ensure?
 - A) Same input produces same output
 - B) Hard to find different input with same hash
 - C) Fast hashing
 - D) Small hash size
7. What is the output size of SHA-512?
 - A) 256 bits
 - B) 128 bits
 - C) 512 bits
 - D) 1024 bits
8. Which technique slows down brute-force attacks?
 - A) Encryption
 - B) Key exchange
 - C) Key stretching
 - D) Compression
9. What is the purpose of a nonce?
 - A) Encryption
 - B) Prevent replay attacks
 - C) Compress data
 - D) Generate keys
10. Which function is irreversible?

- A) Encryption
 - B) Hashing
 - C) Compression
 - D) Encoding
11. Which algorithm is used in DSS?
- A) AES
 - B) DSA
 - C) DES
 - D) RC4
12. What is required to verify a digital signature?
- A) Private key
 - B) Public key
 - C) Session key
 - D) Shared key
13. What does a digital signature guarantee?
- A) Confidentiality
 - B) Integrity and authenticity
 - C) Compression
 - D) Speed
14. What happens if message digest changes?
- A) Signature remains valid
 - B) Verification fails
 - C) Encryption fails
 - D) Speed increases
15. Which attack involves forging a signature?
- A) Replay
 - B) Forgery
 - C) DoS
 - D) Sniffing
16. What is the role of a hash in signatures?
- A) Encrypt data
 - B) Reduce size
 - C) Ensure integrity
 - D) Store keys
17. Which key is kept secret in signatures?
- A) Public key
 - B) Private key
 - C) Session key
 - D) Shared key
18. What ensures trust in certificates?
- A) Encryption
 - B) Certificate Authority
 - C) Compression
 - D) Hash size
19. What is certificate revocation?
- A) Key generation
 - B) Canceling validity
 - C) Encryption
 - D) Compression
20. What is OCSP used for?
- A) Encryption

- B) Certificate validation
 - C) Compression
 - D) Routing
21. What is the main goal of Kerberos tickets?
- A) Encryption
 - B) Authentication
 - C) Compression
 - D) Routing
22. Which protocol uses tickets for authentication?
- A) SSL
 - B) Kerberos
 - C) FTP
 - D) HTTP
23. What is a session key?
- A) Permanent key
 - B) Temporary key
 - C) Public key
 - D) Private key
24. What is mutual authentication?
- A) One-way authentication
 - B) Both parties verify each other
 - C) Anonymous login
 - D) Weak authentication
25. What ensures freshness in Kerberos?
- A) Encryption
 - B) Timestamp
 - C) Compression
 - D) Hashing
26. What is a replay attack?
- A) Data modification
 - B) Resending valid data
 - C) Encryption
 - D) Compression
27. What is the function of an authenticator in Kerberos?
- A) Data storage
 - B) Identity proof
 - C) Encryption
 - D) Compression
28. What happens if system clocks differ in Kerberos?
- A) Faster authentication
 - B) Failure
 - C) Better security
 - D) No effect
29. What is the KDC composed of?
- A) Client & Server
 - B) AS & TGS
 - C) Firewall & IDS
 - D) Router & Switch
30. What is the AS in Kerberos?
- A) Application Server
 - B) Authentication Server

- C) Access Server
 - D) Admin Server
31. What is X.509 certificate format used in?
 - A) Networking
 - B) Digital certificates
 - C) Compression
 - D) Routing
 32. What identifies the certificate owner?
 - A) Public key
 - B) Subject field
 - C) Signature
 - D) Hash
 33. What ensures certificate integrity?
 - A) Encryption
 - B) Digital signature
 - C) Compression
 - D) Routing
 34. What is a root certificate?
 - A) Expired certificate
 - B) Self-signed trusted certificate
 - C) Invalid certificate
 - D) Temporary certificate
 35. What is certificate pinning?
 - A) Storing keys
 - B) Binding certificate to host
 - C) Encryption
 - D) Compression
 36. What is the risk of weak CA validation?
 - A) Faster speed
 - B) MITM attacks
 - C) Compression
 - D) Routing
 37. What is the purpose of trust chain?
 - A) Encryption
 - B) Validate certificate hierarchy
 - C) Compress data
 - D) Routing
 38. What is digital certificate used for?
 - A) Compression
 - B) Authentication
 - C) Routing
 - D) Storage
 39. What ensures non-repudiation?
 - A) Encryption
 - B) Digital signature
 - C) Compression
 - D) Routing
 40. What is a hash collision?
 - A) Same input
 - B) Same output for different inputs
 - C) Encryption
 - D) Compression
 41. Which hash is strongest?

- A) MD5
- B) SHA-1
- C) SHA-256
- D) CRC

42. What is a one-way function?

- A) Reversible
- B) Irreversible
- C) Fast
- D) Small

43. What ensures authentication in PKI?

- A) Encryption
- B) Certificates
- C) Compression
- D) Routing

44. What is key escrow?

- A) Key deletion
- B) Key storage by third party
- C) Encryption
- D) Compression

45. What is identity spoofing?

- A) Encryption
- B) Fake identity
- C) Compression
- D) Routing

46. What is trust anchor?

- A) Root CA
- B) Client
- C) Server
- D) Router

47. What ensures integrity of message?

- A) Encryption
- B) Hash
- C) Compression
- D) Routing

48. What is digital envelope?

- A) Hash
- B) Encrypted message + key
- C) Compression
- D) Routing

49. What is certificate fingerprint?

- A) Encryption
- B) Hash of certificate
- C) Compression
- D) Routing

50. What is PKI used for?

- A) Routing
- B) Secure communication
- C) Compression
- D) Storage

1. Which property ensures the same input always produces the same hash?

Correct Answer: B) Determinism

Explanation:

Determinism is a fundamental property of cryptographic hash functions. It ensures that whenever the same input is provided, the hash function always generates the same output. This consistency is essential for password verification, digital signatures, and data integrity checking. Without determinism, verifying data would not be possible.

2. What is the primary purpose of SHA-3?

Correct Answer: B) Alternative secure hashing standard

Explanation:

SHA-3 is a secure cryptographic hash algorithm developed as an alternative to the SHA-2 family. It provides strong resistance against collision and pre-image attacks. SHA-3 uses the Keccak algorithm and offers enhanced security for modern cryptographic applications.

3. Which attack tries all possible inputs to match a hash?

Correct Answer: B) Brute-force attack

Explanation:

A brute-force attack attempts every possible password or input until the correct hash is found. It requires significant computational power and time. Security mechanisms such as strong passwords, salting, and key stretching reduce the effectiveness of brute-force attacks.

4. What is a salted hash?

Correct Answer: B) Hash with random data added

Explanation:

A salted hash is created by adding a unique random value called a **salt** to the original password before hashing. This ensures that identical passwords produce different hashes, preventing rainbow table attacks and improving password security.

5. Which algorithm is widely used for password hashing?

Correct Answer: C) bcrypt

Explanation:

bcrypt is a password hashing algorithm specifically designed for secure password storage. It automatically includes salting and key stretching, making brute-force attacks much slower. It is widely used in authentication systems and web applications.

6. What does second pre-image resistance ensure?

Correct Answer: B) Hard to find different input with same hash

Explanation:

Second pre-image resistance ensures that given an input and its hash value, it is computationally infeasible to find another different input that produces the same hash. This property protects the integrity of digital data and prevents attackers from replacing valid data with malicious data.

7. What is the output size of SHA-512?

Correct Answer: C) 512 bits

Explanation:

SHA-512 is a member of the SHA-2 family and generates a fixed output of **512 bits (64 bytes)**. Its large hash size provides a high level of security and is commonly used in digital signatures, certificates, and secure communication protocols.

8. Which technique slows down brute-force attacks?

Correct Answer: C) Key stretching

Explanation:

Key stretching increases the time required to generate a password hash by performing thousands of hashing iterations. Algorithms such as PBKDF2, bcrypt, and Argon2 use this technique to make password cracking much more difficult and time-consuming.

9. What is the purpose of a nonce?

Correct Answer: B) Prevent replay attacks

Explanation:

A nonce is a random value that is used only once during an authentication session. It ensures that each communication session is unique and prevents attackers from reusing previously captured messages, thereby protecting against replay attacks.

10. Which function is irreversible?

Correct Answer: B) Hashing

Explanation:

Hashing is a one-way process that converts input data into a fixed-length hash value. Unlike encryption, hashing cannot be reversed to obtain the original data. This property makes hashing suitable for password storage, digital signatures, and data integrity verification.

11. Which algorithm is used in DSS?

Correct Answer: B) DSA

Explanation:

The **Digital Signature Standard (DSS)** is based on the **Digital Signature Algorithm (DSA)**. It is used to generate and verify digital signatures, ensuring the authenticity and integrity of digital documents. DSS does not provide encryption but focuses on authentication and non-repudiation.

12. What is required to verify a digital signature?

Correct Answer: B) Public key

Explanation:

A digital signature is verified using the **sender's public key**. The sender creates the signature using their private key, and the receiver verifies it using the corresponding public key. This process confirms that the message was sent by the legitimate sender and has not been modified.

13. What does a digital signature guarantee?

Correct Answer: B) Integrity and authenticity

Explanation:

A digital signature ensures that the message has not been altered during transmission (**integrity**) and confirms the identity of the sender (**authenticity**). It also supports **non-repudiation**, meaning the sender cannot deny sending the message.

14. What happens if the message digest changes?

Correct Answer: B) Verification fails

Explanation:

If even a single bit of the message changes, the generated hash (message digest) also changes. During signature verification, the receiver compares the calculated digest with the original digest. If they differ, the digital signature verification fails, indicating that the message has been tampered with.

15. Which attack involves forging a signature?

Correct Answer: B) Forgery

Explanation:

A forgery attack occurs when an attacker creates a fake digital signature to impersonate a legitimate sender. Strong cryptographic algorithms such as RSA, DSA, and ECDSA make forging digital signatures computationally infeasible.

16. What is the role of a hash in signatures?

Correct Answer: C) Ensure integrity

Explanation:

Before signing, a hash function generates a fixed-size digest of the message. The digital signature is created using this digest instead of the entire message, making the process more efficient while ensuring that any modification to the message can be detected.

17. Which key is kept secret in signatures?

Correct Answer: B) Private key

Explanation:

The **private key** is known only to the owner and is used to generate digital signatures. The corresponding **public key** is shared with others for signature verification. Keeping the private key confidential is essential for maintaining security.

18. What ensures trust in certificates?

Correct Answer: B) Certificate Authority

Explanation:

A **Certificate Authority (CA)** is a trusted organization that issues and digitally signs X.509 certificates. Users trust certificates because they are verified and issued by a recognized CA, which establishes trust between communicating parties.

19. What is certificate revocation?

Correct Answer: B) Canceling validity

Explanation:

Certificate revocation is the process of invalidating a digital certificate before its expiration date. Certificates may be revoked if the private key is compromised, the certificate holder's information changes, or the certificate is issued incorrectly. Revocation information is distributed through CRLs or OCSP.

20. What is OCSP used for?

Correct Answer: B) Certificate validation

Explanation:

OCSP (Online Certificate Status Protocol) is used to check the current validity of a digital certificate. Instead of downloading an entire Certificate Revocation List (CRL), OCSP provides a real-time response indicating whether a certificate is valid, revoked, or unknown, improving the efficiency of certificate verification.

21. What is the main goal of Kerberos tickets?

Correct Answer: B) Authentication

Explanation:

Kerberos tickets are used to authenticate users securely without transmitting passwords over the network. Once a user is authenticated, the ticket is presented to access network services. This reduces the risk of password theft and improves overall network security.

22. Which protocol uses tickets for authentication?

Correct Answer: B) Kerberos

Explanation:

Kerberos is a network authentication protocol that uses tickets to verify the identity of users and services. It provides secure authentication using secret-key cryptography and prevents unauthorized access in distributed systems.

23. What is a session key?

Correct Answer: B) Temporary key

Explanation:

A session key is a temporary cryptographic key generated for a single communication session. It encrypts data exchanged between the client and the server during that session. After the session ends, the key is discarded, improving security.

24. What is mutual authentication?

Correct Answer: B) Both parties verify each other

Explanation:

Mutual authentication is a security process where both the client and the server authenticate each other's identities before communication begins. This prevents impersonation attacks and ensures that both parties are legitimate.

25. What ensures freshness in Kerberos?

Correct Answer: B) Timestamp

Explanation:

Kerberos uses **timestamps** to ensure that authentication requests are recent and have not been replayed. If a timestamp is outside the allowed time window, the request is rejected. This mechanism helps prevent replay attacks.

26. What is a replay attack?

Correct Answer: B) Resending valid data

Explanation:

A replay attack occurs when an attacker captures a valid authentication message and retransmits it later to gain unauthorized access. Security mechanisms such as timestamps and nonces are used to detect and prevent replay attacks.

27. What is the function of an authenticator in Kerberos?

Correct Answer: B) Identity proof

Explanation:

An authenticator is a data structure sent by the client to prove its identity during authentication. It contains the client's identity and a timestamp encrypted with the session key, ensuring that the request is genuine and fresh.

28. What happens if system clocks differ in Kerberos?

Correct Answer: B) Failure

Explanation:

Kerberos relies on synchronized clocks between clients and servers. If the system clocks differ significantly, timestamps cannot be verified correctly, causing authentication to fail. Therefore, time synchronization is essential in Kerberos environments.

29. What is the KDC composed of?

Correct Answer: B) AS & TGS

Explanation:

The **Key Distribution Center (KDC)** is the trusted third party in Kerberos. It consists of two components:

- **Authentication Server (AS)** – Verifies the user's identity.
 - **Ticket Granting Server (TGS)** – Issues service tickets after successful authentication.
- Together, they enable secure access to network services.

30. What is the AS in Kerberos?

Correct Answer: B) Authentication Server

Explanation:

The **Authentication Server (AS)** is the first component of the Kerberos protocol. It verifies the user's credentials and issues a **Ticket Granting Ticket (TGT)**. The client then uses this TGT to request service tickets from the Ticket Granting Server (TGS), enabling secure authentication without repeatedly sending passwords.

31. What is X.509 certificate format used in?

Correct Answer: B) Digital certificates

Explanation:

The **X.509** standard defines the format of digital certificates used in Public Key Infrastructure (PKI). These certificates contain information such as the subject's identity, public key, issuer details, validity period, and digital signature. X.509 certificates are widely used in SSL/TLS, secure email, and digital authentication.

32. What identifies the certificate owner?

Correct Answer: B) Subject field

Explanation:

The **Subject** field in an X.509 certificate contains the identity of the certificate owner. It includes information such as the person's or organization's name, domain name, and other identification details. This field helps users verify the identity of the certificate holder.

33. What ensures certificate integrity?

Correct Answer: B) Digital signature

Explanation:

A Certificate Authority (CA) digitally signs every X.509 certificate. The digital signature ensures that the certificate has not been modified after issuance. If any information in the certificate is changed, the signature verification fails, indicating that the certificate is no longer trustworthy.

34. What is a root certificate?

Correct Answer: B) Self-signed trusted certificate

Explanation:

A **Root Certificate** is a self-signed certificate issued by a trusted Root Certificate Authority (CA). It serves as the foundation of the certificate trust chain. Operating systems and web browsers maintain a list of trusted root certificates to verify the authenticity of other certificates.

35. What is certificate pinning?

Correct Answer: B) Binding certificate to host

Explanation:

Certificate pinning is a security technique that associates a specific certificate or public key with a particular website or application. During future connections, the application verifies that the same certificate is presented, helping to prevent Man-in-the-Middle (MITM) attacks.

36. What is the risk of weak CA validation?

Correct Answer: B) MITM attacks

Explanation:

If Certificate Authorities (CAs) are not properly validated or trusted, attackers may use fake certificates to impersonate legitimate websites. This can lead to **Man-in-the-Middle (MITM) attacks**, where attackers intercept and manipulate communication between two parties.

37. What is the purpose of trust chain?

Correct Answer: B) Validate certificate hierarchy

Explanation:

The certificate trust chain establishes trust by verifying each certificate from the **End-Entity Certificate** through the **Intermediate CA** to the **Root CA**. If every certificate in the chain is valid and trusted, the communication is considered secure.

38. What is digital certificate used for?

Correct Answer: B) Authentication

Explanation:

A digital certificate is used to authenticate the identity of users, servers, or organizations. It confirms that the public key belongs to the claimed entity, enabling secure communication over networks such as the Internet.

39. What ensures non-repudiation?

Correct Answer: B) Digital signature

Explanation:

A digital signature provides **non-repudiation**, meaning the sender cannot deny sending the message. Since the signature is created using the sender's private key, only the

legitimate owner can generate it, making the sender accountable for the transmitted data.

40. What is a hash collision?

Correct Answer: B) Same output for different inputs

Explanation:

A hash collision occurs when two different inputs produce the same hash value. Cryptographic hash functions such as SHA-256 are specifically designed to make finding collisions computationally infeasible, ensuring data integrity and security.

41. Which hash is strongest?

Correct Answer: C) SHA-256

Explanation:

SHA-256 is a member of the SHA-2 family and is considered one of the strongest and most widely used cryptographic hash algorithms. It generates a 256-bit hash value and provides strong resistance against collision and pre-image attacks. It is commonly used in digital signatures, SSL/TLS, and blockchain technology.

42. What is a one-way function?

Correct Answer: B) Irreversible

Explanation:

A one-way function is a mathematical function that is easy to compute but extremely difficult to reverse. Cryptographic hash functions are one-way because the original input cannot be reconstructed from the hash value. This property makes them ideal for password storage and data integrity verification.

43. What ensures authentication in PKI?

Correct Answer: B) Certificates

Explanation:

In a **Public Key Infrastructure (PKI)**, digital certificates provide authentication by binding a user's identity to a public key. These certificates are issued and digitally signed by trusted Certificate Authorities (CAs), allowing users to verify the identity of communicating parties.

44. What is key escrow?

Correct Answer: B) Key storage by third party

Explanation:

Key escrow is the practice of storing cryptographic keys with a trusted third party. This allows authorized organizations to recover encrypted data if the original key is lost or unavailable. While useful for recovery, it must be managed securely to prevent unauthorized access.

45. What is identity spoofing?

Correct Answer: B) Fake identity

Explanation:

Identity spoofing is an attack in which an attacker pretends to be another user or device to gain unauthorized access. Techniques such as strong authentication, digital certificates, and multi-factor authentication help prevent identity spoofing.

46. What is a trust anchor?

Correct Answer: A) Root CA

Explanation:

A **Trust Anchor** is the trusted **Root Certificate Authority (Root CA)** that serves as the starting point for certificate validation. Systems and web browsers maintain trusted Root CA certificates to verify the authenticity of certificate chains.

47. What ensures integrity of message?

Correct Answer: B) Hash

Explanation:

A cryptographic hash function ensures message integrity by generating a unique digest for the original data. If even one bit of the message changes, the resulting hash value changes completely. Comparing hash values helps detect unauthorized modifications.

48. What is digital envelope?

Correct Answer: B) Encrypted message + key

Explanation:

A digital envelope is a secure method of transmitting encrypted data. The message is encrypted using a symmetric session key, and the session key is then encrypted using the recipient's public key. This combines the speed of symmetric encryption with the security of asymmetric encryption.

49. What is certificate fingerprint?

Correct Answer: B) Hash of certificate

Explanation:

A certificate fingerprint is a cryptographic hash value (such as SHA-256) calculated from the contents of a digital certificate. It uniquely identifies the certificate and allows users to verify that it has not been altered or replaced.

50. What is PKI used for?

Correct Answer: B) Secure communication

Explanation:

Public Key Infrastructure (PKI) is a framework that manages digital certificates and public-key encryption. It enables secure communication by providing authentication, encryption, digital signatures, and certificate management. PKI is widely used in HTTPS, secure email, VPNs, and e-commerce applications.